electric motor horsepower

ELECTRIC MOTOR HORSEPOWER REQUIRED TO DRIVE A HYDRAULIC PUMP

GPM	100 PSI	200 PSI	250 PSI	300 PSI	400 PSI	500 PSI	750 PSI	1000 PSI	1250 PSI	1500 PSI	2000 PSI	2500 PSI	3000 PSI
1/2	.04	.07	.09	.11	.14	.18	.26	.35	.44	.53	.70	.88	1.10
1	.07	.14	.18	.21	.28	.35	.52	.70	.88	1.05	1.40	1.76	1.92
1-1/2	.10	.21	.26	.31	.41	.52	.77	1.03	1.29	1.55	2.06	2.58	3.09
2	.14	.28	.35	.42	.56	.70	1.04	1.40	1.76	2.10	2.80	3.53	4.20
2-1/2	.17	.34	.43	.51	.69	.86	1.29	1.72	2.15	2.58	3.44	4.30	5.14
3	.21	.42	.53	.63	.84	1.05	1.56	2.10	2.64	3.15	4.20	5.28	6.30
3-1/2	.24	.48	.60	.72	.96	1.20	1.80	2.40	3.00	3.60	4.80	6.00	7.20
4	.28	.56	.70	.84	1.12	1.40	2.08	2.80	3.52	4.20	5.60	7.04	8.40
5	.35	.70	.88	1.05	1.40	1.75	2.60	3.50	4.40	5.25	7.00	8.80	10.50
6	.42	.84	1.05	1.26	1.68	2.10	3.12	4.20	5.28	6.30	8.40	10.56	12.60
7	.49	.98	1.23	1.47	1.96	2.45	3.64	4.90	6.16	7.35	9.80	12.32	14.70
8	.56	1.12	1.40	1.68	2.24	2.80	4.16	5.60	7.04	8.40	11.20	14.08	16.80
9	.62	1.24	1.55	1.86	2.48	3.10	4.65	6.18	7.73	9.28	12.40	15.56	18.58
10	.70	1.40	1.75	2.10	2.80	3.50	5.20	7.00	8.80	10.50	14.00	17.60	21.00
11	.77	1.54	1.93	2.31	3.08	3.85	5.72	7.70	9.68	11.50	15.40	19.36	23.10
12	.84	1.68	2.10	2.52	3.36	4.20	6.24	8.40	10.50	12.60	16.80	21.00	25.20
13	.89	1.78	2.23	2.67	3.56	4.45	6.68	8.92	11.20	13.40	17.80	22.40	26.72
14	.96	1.92	2.40	2.88	3.84	4.80	7.20	9.60	12.00	14.40	19.20	24.00	28.80
15	1.05	2.10	2.63	3.15	4.20	5.25	7.80	10.50	13.20	15.70	21.00	26.40	31.50
16	1.10	2.20	2.75	3.30	4.40	5.50	8.25	11.00	13.80	16.50	22.00	27.60	33.00
17	1.17	2.34	2.93	3.51	4.68	5.85	8.78	11.70	14.60	17.60	23.40	29.20	35.10
18	1.26	2.52	3.15	3.78	5.04	6.30	9.35	12.60	15.80	18.90	25.20	31.60	37.80
19	1.30	2.60	3.25	3.90	5.20	6.50	9.75	13.00	16.30	19.50	26.00	32.60	39.00
20	1.40	2.80	3.50	4.20	5.60	7.00	10.40	14.00	17.60	21.00	28.00	35.20	42.00
25	1.75	3.50	4.38	5.25	7.00	8.75	13.10	17.50	21.90	26.20	35.00	43.80	52.50
30	2.10	4.20	5.25	6.30	8.40	10.50	15.60	21.00	26.40	31.50	42.00	52.80	63.00
35	2.45	4.90	6.13	7.35	9.80	12.20	18.40	24.50	30.60	36.70	49.00	61.20	73.50
40	2.80	5.60	7.00	8.40	11.20	14.00	20.80	28.00	35.20	42.00	56.00	70.40	84.00
45	3.15	6.30	7.87	9.45	12.60	15.80	23.60	31.50	39.40	47.30	63.00	78.80	94.50
50	3.50	7.00	8.75	10.50	14.00	17.50	26.00	35.00	44.00	52.50	70.00	88.00	105.00
55	3.85	7.70	9.63	11.60	15.40	19.30	28.60	38.50	48.40	57.80	77.00	96.80	115.50
60	4.20	8.40	10.50	12.60	16.80	21.00	31.20	42.00	52.80	63.00	84.00	105.60	126.00
65	4.55	9.10	11.40	13.60	18.20	22.80	33.80	45.50	57.20	68.20	90.00	114.40	136.50

table c-1

The **table c-1,** above is based on a pump efficiency of 85%, and is calculated from the formula:

HP = GPM x PSI ÷ (1714 x .85)

As horsepower varies directly with flow or pressure, multiply proportionately to determine values not shown. For example, at 4000 PSI, multiply 2000 PSI values by 2.

standard enclosures for electric motors

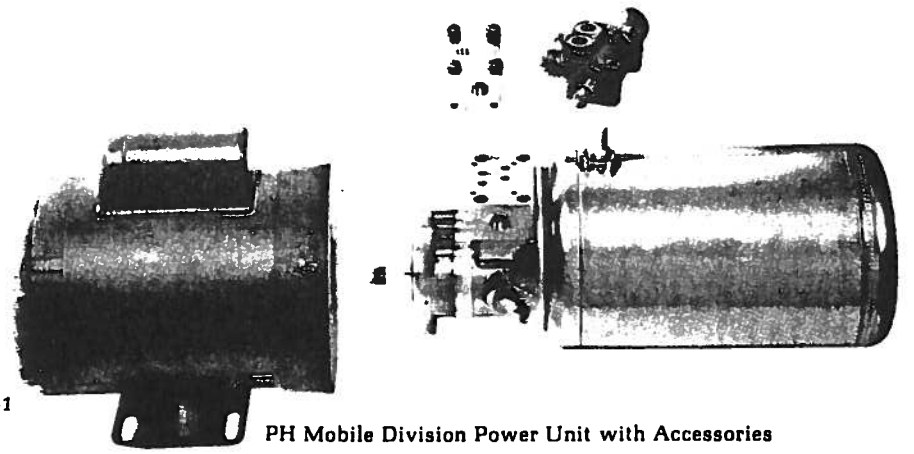


illustration c-1 PH Mobile Division Power Unit with Accessories

Open
The open motor is one having ventilating openings which permit passage of external cooling air over and around the windings.

Drip-Proof
The drip-proof motor is an open motor in which ventilating openings are so constructed that drops of liquid or solids falling on the machine at any angle not greater than 15 degrees from the vertical cannot enter the machine.

Guarded
A guarded motor is an open motor in which ventilating openings are limited to specified size and shape to prevent insertion of fingers or rods to avoid accidental contact with rotating or electrical parts.

Splash-Proof
A splash-proof motor is an open motor in which ventilating openings are so constructed that drops of liquid or solid particles falling on the machine or coming toward the machine in a straight line at any angle not greater than 100 degrees from the vertical cannot enter the machine.

Totally-Enclosed
A totally-enclosed motor is a motor so enclosed as to prevent the free exchange of air between the inside and outside of the case, but not air-tight.

Totally-Enclosed Nonventilated (TENV)
A totally-enclosed nonventilated (TENV) motor is a totally-enclosed motor which is not equipped for cooling by means external to the enclosing parts.

Totally-Enclosed Fan-Cooled (TEFC)
A totally-enclosed fan-cooled (TEFC) motor is a totally enclosed motor with a fan to blow cooling air across the external frame. It is a popular motor for use in dusty, dirty, and corrosive atmospheres.

Encapsulated
Encapsulated motor is an open motor in which the windings are covered with a heavy coating of material to protect them from moisture, dirt, abrasion, etc. Some encapsulated motors have only the coil noses coated. In others, the encapsulation material impregnates the windings even in the coil slots. With this complete protection, the motors can often be used in applications which formerly demand totally enclosed motors.

Explosion-Proof
An explosion-proof motor is a totally enclosed motor designed and built to withstand an explosion of gas or vapor within it, and to prevent ignition of gas or vapor surrounding the machine by sparks, flashes or explosions which may occur within the machine casing.

three-phase motor design

Design "B" — A Design "B" motor is a 3-phase squirrel-cage motor designed to withstand full-voltage starting and developing lock-rotor and breakdown torques adequate for general application.

Design "C" — A Design "C" motor is a 3-phase squirrel-cage motor designed to withstand full-voltage starting, developing locked-rotor torque for special high torque applications.

Design "D" — A Design "D" motor is a 3-phase squirrel-cage motor designed to withstand full-voltage starting, developing 275 percent locked-rotor torque (generally referred to as a "high slip" motor).

mounting flanges continued

pump mounting bracket adapts following Parker Pumps — To 1/4 HP thru 20 HP Motors

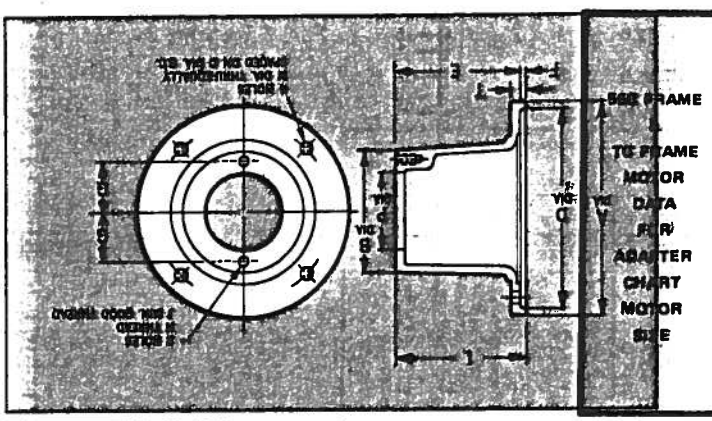
D, H & M Series Gear Pumps
PVV 23/33 Variable Volume, Pressure Compensated, Vane Pumps
(and other pumps with similar mounting flanges)

Coupling Installation — The Parker totally enclosed pump mounting bracket offers excellent noise absorbtion and safety from the rotating shafts and coupling. The bracket is designed to mount on the motor face with the motor coupling half secure to the shaft. Then the pump, with its coupling half secure on the pump shaft, is mounted and the coupling halves are engaged. This will require proper spacing of the coupling prior to installation and a coupling with an outside diameter less than "P" dimension. If the coupling selected cannot be assembled this way, both coupling halves must be installed on the motor shaft. Next, mount the adapter on the motor. Then the pump can be mounted and the coupling secured to the pump by using the access slot to tighten the pump shaft coupling set screw.

ADAPTER MODEL SELECTION CHART

ELECTRIC MOTOR SIZE		SAE AA MTG		STD SHAFT SAE A MTG		LONG SHAFT		SAE B MTG		SAE C MTG	
		D GEAR		H GEAR				M GEAR PW23/33 VAR VANE VPO33 VAR PISTON PAF 10/16/20/25 PISTON PAV 16 VAR PISTON		VPO58 VAR PISTON PAF 32/40/50 PISTON PAV 32 VAR PISTON	
FRAME	SHAFT	ADAPTER MODEL	REC CPLG	ADAPTER MODEL	REC CPLG	ADAPTER MODEL	REC CPLG	ADAPTER MODEL	REC CPLG		
56C	5/8	DS	L-075 LOVEJOY	HS	L-075 LOVEJOY						
143TC	7/8		L-095 LOVEJOY		L-095 LOVEJOY						
145TC											
182TC	1-1/8	DM	AL-100 LOVEJOY	HM	AL-100 LOVEJOY	HML	AL-110 LOVEJOY	MM	AL-100 LOVEJOY	ADAPTER MODEL	REC COUPLING
184TC											
213TC	1-3/6		300 MAGNALOY		300 MAGNALOY		AL-110U LOVEJOY			RM	AL-110 LOVEJOY
215TC											
254TC	1-5/8				400 MAGNALOY				400 MAGNALOY		400 MAGNALOY
256TC											
284TC	1-7/8							MM + RING 753006	AL-150 LOVEJOY	RM + RING 753006	AL-150 LOVEJOY
286TC											

table i-4



TYPE	OPDP	OPDP OR TEFC		FRAME
RPM	3600	1800	1200	SIZE
HP	½–1	¼–½		56C
	2	1	½	143TC
	3	2	1	145TC
	5	3	1½	182TC
	7½	5	2	184TC
	10	7½	3	213TC
	15	10	5	215TC
	20	15	7½	254TC
	25	20	10	256TC
		25	15	284TC
		30	20	286TC

MODEL	A	B	C	D	E	F	G	H	J	L	M	P	T	WEIGHT
DS	6.5	4.5	5.88	4.50	4.04	.21	1.62	3/8-16	Thru	4.25	.44	2.00	.50	8 lbs.
HS	6.5	4.6	5.88	4.50	4.04	.21	2.09	3/8-16	.75	4.25	.44	3.25	.50	7 lbs.
DM	9.0	5.1	7.25	8.50	5.25	.13	1.62	3/8-16	Thru	5.38	.53	2.00	.50	13 lbs.
HM	9.0	5.1	7.25	8.50	5.25	.13	2.09	3/8-16	.88	5.38	.53	3.25	.50	12 lbs.
HML	9.0	5.1	7.25	8.50	6.75	.13	2.09	3/8-16	.88	6.88	.53	3.25	.56	14 lbs.
MM	9.0	6.5	7.25	8.50	6.25	.13	2.88	1/2-13	1.00	6.38	.53	4.00	.50	14 lbs.
RM	9.0	8.38	7.25	8.50	6.69	.13	3.56	5/8-11	Thru	6.82	.56	5.00	.63	17 lbs.

table i-5

pump selection factors

Pressure Capability — One of the most critical factors affecting selection of a pump is its pressure rating. Where there is any doubt concerning the peak pressure to which a pump will be exposed, the pump should be slightly overspecified.

Some personnel often have no idea at what pressure a pump is operating because of the absence of pressure indicators. As a result, pumps often operate at excessive pressure levels. Over-pressure operation results in accelerated wear and ultimate early failure. This problem is common in the industry. To avoid it, systems should be equipped with a remote gage or with a simple means for inserting a gage to perform maintenance checks.

Relief valves are used with fixed-displacement pumps to maintain a fixed pressure setting when the system flow requirement is below the pump output. Plant personnel frequently will adjust this valve to speed up cycle time or to lift a heavier load. This action can result in the pump's rated pressure being exceeded. One solution is to insert "once-adjusted" or sealed relief valves.

Pump Selection — Selection of a pump starts with the selection of a working pressure for the system. Working pressure reflects system cost. A 500-p.s.i. system requires a larger line, larger components, and a larger pump to perform a job than does a 2,000-p.s.i. system. The 2,000-p.s.i. system may have lower initial cost, but if it is not properly engineered, it may require more maintenance. The tendency today, with better maintenance and better components available, is for designers to specify systems with higher pressures to achieve more work at a lower cost.

Once the system pressure has been established, the pump selection can be made. The designer should consider the function (variable or fixed), maximum pressure recommended by the manufacturer, volume, cost, noise, and reliability.

Drive Speed — Excessive operating speeds often result in pump cavitation, which occurs when the suction exceeds a limit the fluid cannot tolerate. Under these conditions bubbles form in the fluid. When these bubbles are exposed to the pressure cavity, they collapse rapidly. This action is called cavitation. It results in noise, erosion, and eventual destruction of the pump.

Essentially, two drive speeds are utilized in industrial pump applications—1,800 and 1,200 rpm. It is recommended that the higher speed be utilized where possible since a greater horsepower-per-dollar value can be achieved at the higher speeds in systems with shaft inputs up to 100 hp. Some pumps are capable of speeds up to 3,600 rpm, but at this speed a noise problem may be encountered, and the danger of cavitation is increased.

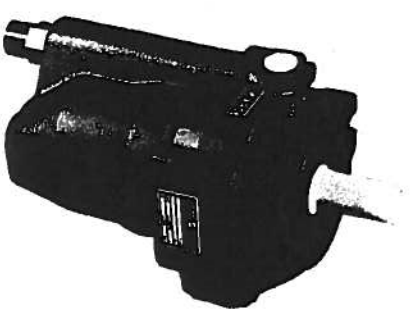
To avoid cavitation the pump suction line should (1) be as short as possible, (2) have as large a diameter as practicable, and (3) have a minimum number of elbows.

Environment — The pump is the most critical element in a hydraulic circuit relative to environment. In metal casting applications, for example, the pump will be the first to wear out due to contamination. In general, a system that will provide sufficient protection for the pump also will provide more than adequate protection for all of the other elements.

Filters must be selected carefully to keep contaminants out of the system. A system without a filter is asking for trouble. Therefore a designer should be generous in selecting a filter. However, no matter how good the filter, some foreign materials will find their way into the system. The best that can be achieved is to minimize contaminants.

Reservoir design also is an important factor in cutting down on system contamination. The designer should arrange the breather and fill locations so that they do not collect dirt particles.

Environmental temperatures also affect operation. Some foundries experience extremes in temperature in the molding area. The temperature can go down to 35°F in the winter and up to 110°F in the summer. These variations cause the fluid to undergo viscosity changes that may result in (1) pump cavitation due to high viscosity at low temperatures or (2) excessive internal slippage due to low viscosity at high temperatures. Fluid selection in such



PH Pressure Compensated Pump

illustration i-1

pump selection factors continued

situations is critical and may require different types during different seasons.

System Fluids — Although hydraulic fluids do not readily wear out or break down, they should be changed periodically to assure that clean fluid is in the system at all times. Additives in the fluid do break down, and even may be filtered out by a good filtering system. Fluids with anti-wear additives and foam depressants now are common, and they should be specified. Fire resistance of fluids is another consideration. Recommendations of both pump manufacturers and fluid suppliers should be studied carefully before a system is put into operation.

notes

hydraulic pump calculation formula

Hp	= Horsepower	V	= Speed (in/min)
RPM	= Revolutions per minute	π	= 3.14
w	= Work	r	= Radius
d	= Displacement – cubic inches	psi.	= Pounds per square inch
Q	= Flow rate – gpm	PSIA	= Pounds per square inch-absolute
G.P.M.	= Gallons per minute (U.S.)	PSIG	= Pounds per square inch-gauge
T	= Torque	ΔP	= Pressure drop – p.s.i.
Sp.Gr.	= Specific gravity	231 $in.^3$	= 1 U.S. Gallon
F	= Force (pounds)		

PUMP OUTPUT FLOW

$$1 \text{ GAL.} = 231 \frac{in^3}{}$$

$$\text{Theoretical output (gpm)} = \frac{\text{Displacement (cu. in.)/rev. x rpm}}{\text{231 cu. in.}}$$

$$\text{G.P.M.} = \frac{\text{RPM x d}}{231}$$

VOLUMETRIC EFFICIENCY PERCENT

$$\text{volumetric efficiency (pump)} \quad \frac{\text{actual displacement (G.P.M.)}}{\text{theoretical displacement (G.P.M.)}} \times 100$$

MECHANICAL EFFICIENCY PERCENT

$$\text{\% Mech. eff.} = \frac{\text{Theoretical torque required to drive the pump (lb. in.)}}{\text{Actual torque supplied to pump (lb. in.)}} \times 100$$

OVERALL EFFICIENCY PERCENT

$$\text{Pump over-all eff. \%} = \frac{\text{Output hyd. Hp}}{\text{Input Hp to pump}} \times 100$$

PUMP OUTPUT HORSEPOWER

$$Hp = \frac{Fxv}{396000}$$

PUMP

$$Hp = \frac{\text{G.P.M. x PSI}}{\text{1714 x eff. (pump)}}$$

$$\text{or } Hp = \frac{.000583 \text{ x G.P.M. x p.s.i.}}{\text{eff. (pump)}}$$

$$Hp = \frac{T \text{ x RPM}}{5252} \quad \text{Torque in lb. ft.}$$

$$Hp = \frac{T \text{ x RPM}}{63025} \quad \text{Torque in lb. in.}$$

$$Hp = \frac{\text{RPM } \Delta \text{ Pd } (in^3/\text{rev})}{63025 \ 2\pi}$$

$$Hp = \frac{T \text{ x RPM x } 2\pi}{33{,}000 \text{ x } 12} \quad \text{Torque in lb.-in.}$$

$$Hp = \frac{\text{volts x amperes}}{745.7}$$

POWER

$$\text{power} = \frac{\text{work}}{\text{time}} = \frac{\text{force x distance}}{\text{time}}$$

$$\text{power} = \frac{231 \text{ x Q x p.s.i.}}{12} \quad \frac{\text{ft. - lb.}}{\text{min.}}$$

$$HP = \frac{\text{power}}{33{,}000 \frac{\text{ft. - lbs.}}{\text{min.}}}$$

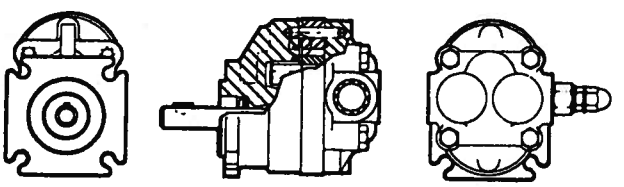
hydraulic pump calculation

hydraulic pump calculation

Pump Pressure Ratings — A 2000 p.s.i. pressure rating does not mean the pump will just develop 2000 p.s.i. in your system. This refers to the maximum pressure which should be developed by the pump allowing a safety factor, without danger of damage to the pumping element over an extended period of time.

By properly setting the system relief valve, you may limit maximum system pressure to any safe value less than the maximum pump rating.

Caution: *A relief valve built into the pump casing should not be used for the main circuit relief valve (other than hand pumps) unless it relieves back into the inlet of the pump, severe local over-heating of the oil can occur in less than one minute. These built-in relief valves are intended only for emergency protection, and should be set higher than the main relief valve setting.*



PH 4 Bolt Flange Gear Pump with Built-in Relief Valve
illustration i-2

Select a pump and drive motor to create about 10 to 25 % greater system pressure than needed at the actuator. This is required to take care of pressure losses in valves, piping and friction of the cylinder rod and piston.

sizing pumps

Example of the pumps displacement vs cylinder travel relationship is:

Example: *A cylinder having a 3-1/4" diameter piston is required to stroke 6 inches. By using a hand pump with a displacement of 1.75 cubic inches per pump stroke (pumping oil during the forward pump stroke only, the return stroke in this case is used to refill the pumping chamber) how many pumping strokes are required to move the cylinder piston 6 inches.*

The area of a 3-1/4" diameter piston is 8.29 square inches. The amount of oil required to move the piston 6 inches is 8.29 inches^2 × 6 inches = 49.6 inches^3. The handpump, having a displacement of 1.75 inches^3 during our pumping stroke and calculating the number of pumping strokes required is 49.6 inches^3 divided by 1.75 inches^3 = 28.4 pumping strokes.

Example: *How many GPM is required to move a 5 inch diameter cylinder piston at the rate of 2 inches per second.*

To calculate how much oil (in gallons per minute) is required to move the piston 2 inches every second

Use the formula:

$$\text{VOL (G.P.M.)} = \frac{A \times L \times 60}{231 \times \text{time in sec.}}$$

Where:

A	=	5 inch dia. piston = 19.6 in.^{2}
L	=	Length = 2 inches
231	=	number of cubic inches/gallon
time in sec.	=	1
60	=	60 sec./min.

The formula is now:

$$\text{VOL (G.P.M.)} = \frac{19.6 \times 2 \times 60}{231 \times 1}$$

$$\frac{19.6 \times 2 \times 60}{231 \times 1} = \frac{2345}{231}$$

$$2345 \div 231 = 10.1 \text{ G.P.M.}$$

Example: *What size pump, rated in GPM, is needed to extend and retract a 5 inch bore,18 inch stroke cylinder. This cylinder has a 2 inch diameter piston rod and the total cycle time (out and back) is 12 seconds.*

Solution:

Using the formula:

$$\text{VOL (G.P.M.)} = \frac{A \times L \times 60}{231 \times \text{time in sec.}}$$

Find the area of a 5 inch Bore cylinder, the Bore = 19.6 inches^2. Find the effective area on the return stroke (this is the piston area less the rod area for the return stroke). This is 19.6 inches^2 - 3.1 inches^2 = the piston rod area 16.5 inches^2 which is the effective working area on the return stroke. We are looking for total cycle time extending and retracting add 19.6 inches^2 + 16.5 inches^2 = 36.1 inches^2 total area. Total motion of the cylinder is 18 inches.

hydraulic pump calculation continued

stroke 18 inches extending + 18 inches retracting equals 36 inches (total travel).

$$\text{VOL (G.P.M.)} = \frac{36.1 \text{ inches}^2 \times 18 \text{ inches} \times 60}{231 \text{ inches}^3 \times 12 \text{ seconds}}$$

$$\frac{36.1 \times 18 \times 60}{231 \times 12} = \frac{38988}{2773}$$

$$38988 \div 2773 = 14. \text{ G.P.M.}$$

When a pump is operating in a system, this pump will develop only the amount of pressure required to do the work. For example: when a cylinder is extending and moving a load, the only pressure developed by the pump and noted on the pressure gauge is what is required to overcome the load and friction. Now when the cylinder piston reaches the opposite end of the cylinder and stalls, the pressure in the system can reach an infinite value. When this happens, and if the system relief valve is left out or failed to function properly, the drive motor will stall, a hose, pipe or tube would burst, or possibly the pump or valve casting would burst. For this reason a system relief valve is required in a system, and set at a pressure setting which is equal to or below the lowest pressure rated component in the system. The system relief should be plumbed as close to the pump as possible. A relief or pressure reducing valve can be used in other parts of the system to limit these componenents to a lower pressure.

notes